



Edexcel IGCSE Physics



Your notes

Forces, Movement & Changing Shape

Contents

- * Types of Forces
- * Scalar & Vectors
- * Resultant Forces
- * Unbalanced Forces
- * Weight
- * Stopping Distance
- * Terminal Velocity
- * Core Practical: Investigating Force & Extension
- * Hooke's Law



Your notes

Types of Forces

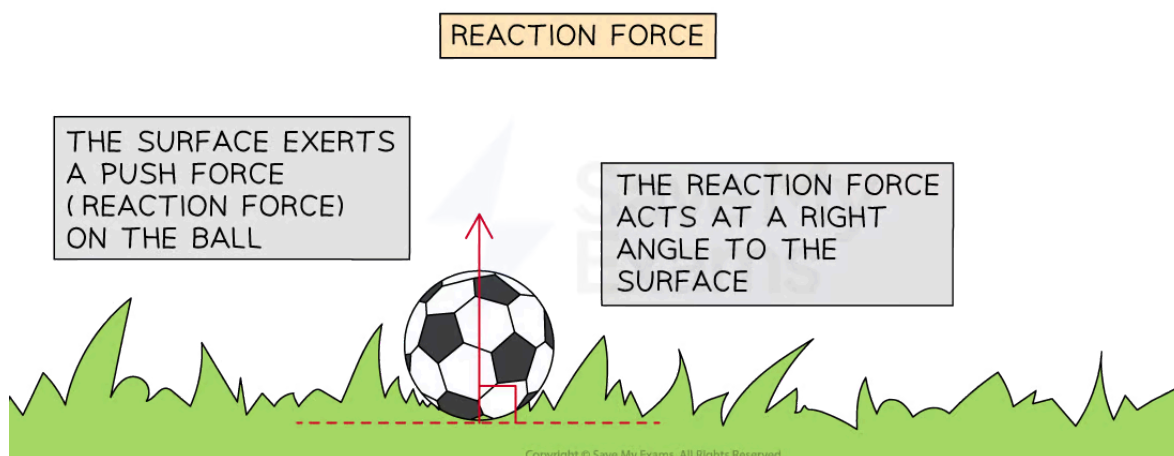
Types of forces

- A **force** is a **push** or **pull** that arises from the **interaction** between objects
- There are lots of different types of forces that occur when objects interact

What are the different types of forces?

- **Gravitational force (weight)**
 - There is a **gravitational force** of attraction between all objects with **mass**
 - The more massive the object, the greater the gravitational force exerted by it
 - When a football is thrown (or kicked), the gravitational pull of the Earth on the football pulls it toward the (centre of the) Earth
- **Reaction force**
 - When an object rests on a surface, the **surface** exerts a push force on the **object**
 - This **reaction** force acts at **right angles** (perpendicular) to the surface
 - When a football rests on the horizontal surface of the grass, the grass exerts a push force (reaction force) vertically upwards on the football

Reaction force acting on a football



The push force exerted on an object by a surface is called the reaction force

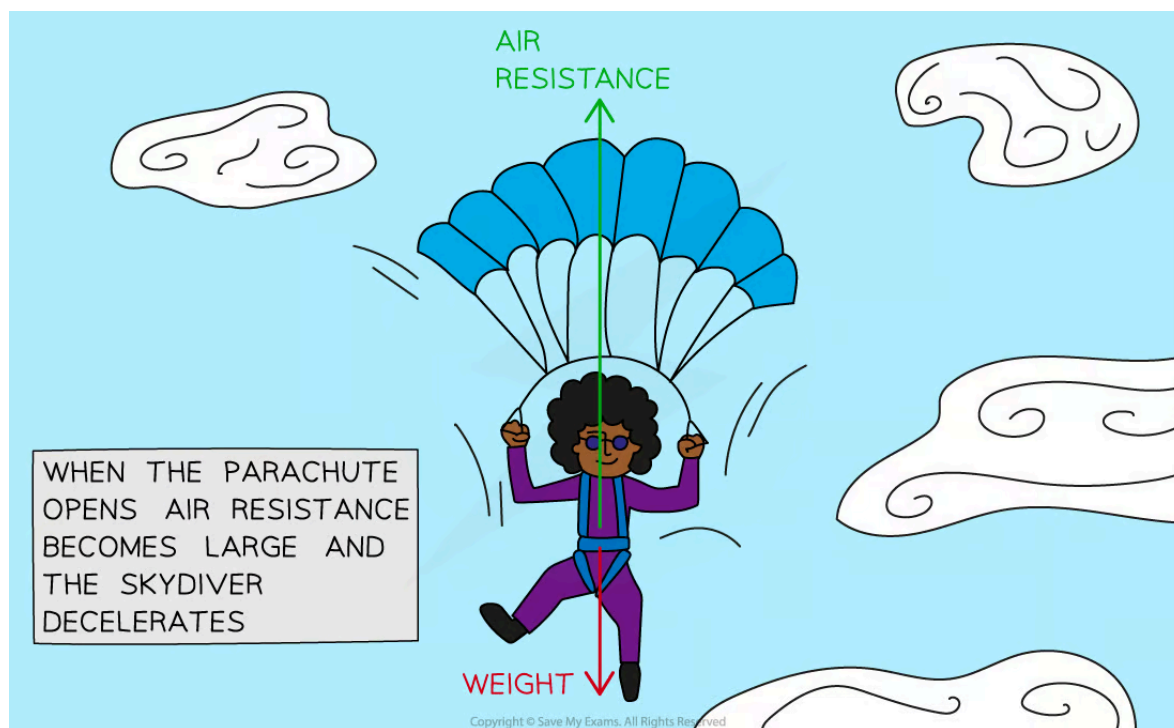
- **Friction**
 - Frictional forces always **oppose** the **motion** of an object, causing it to **slow down**
 - Friction occurs when two **surfaces** move over one another
 - When a box is pushed across a carpet, the carpet exerts a frictional force on the box, slowing its motion
- **Drag force**



Your notes

- Drag force is a type of **frictional force** that occurs when an object moves through a **fluid** (a gas or a liquid)
- The particles in the fluid **collide** with the object moving through it and **slow** its **motion**
- When a pebble is thrown into water, the water molecules flow against its solid surface, causing it to slow down
- **Air resistance**
 - Air resistance is a specific type of **drag** force and is therefore also a **frictional** force
 - Air resistance occurs when particles of air **collide** with an object moving through it and **slows** its **motion**
 - When a skydiver opens their parachute, air resistance opposes their motion and reduces their speed so it is safe to land

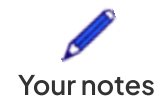
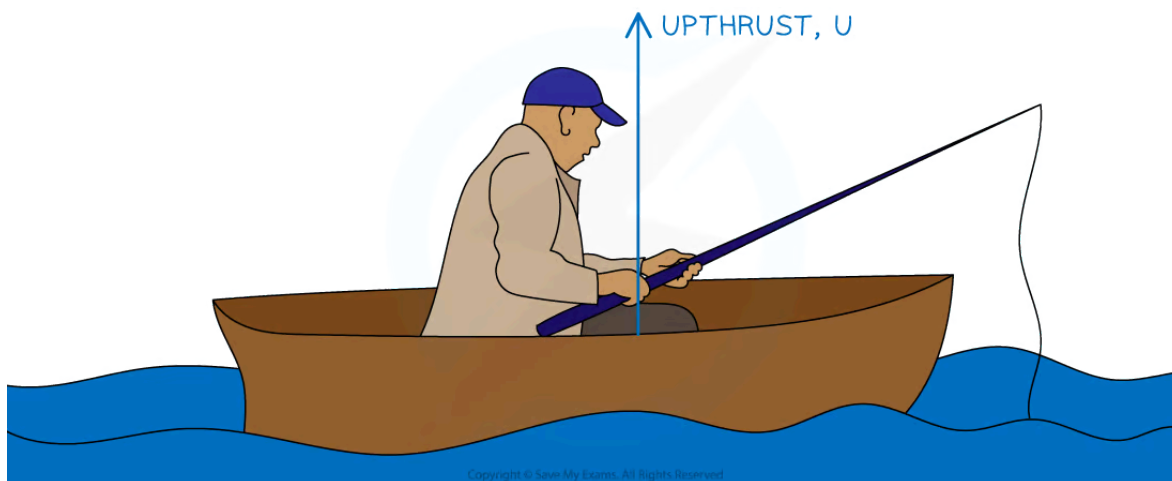
Air resistance acting on a skydiver



A skydiver uses air resistance to reduce their speed so that they can land safely

- **Thrust**
 - Thrust is a force produced by an engine that **speeds up** the motion of an object
 - The engine of a car exerts a thrust force and increases its speed
- **Upthrust**
 - When an object is fully or partially **submerged** in a **fluid**, the fluid exerts an **upward-acting** push force on the object
 - A boat floats on a lake due to the **upthrust** exerted by the water on the boat
 - A ball held underwater will shoot upwards when released due to the upthrust exerted by the water pushing it back to the surface

Upthrust acting on a boat

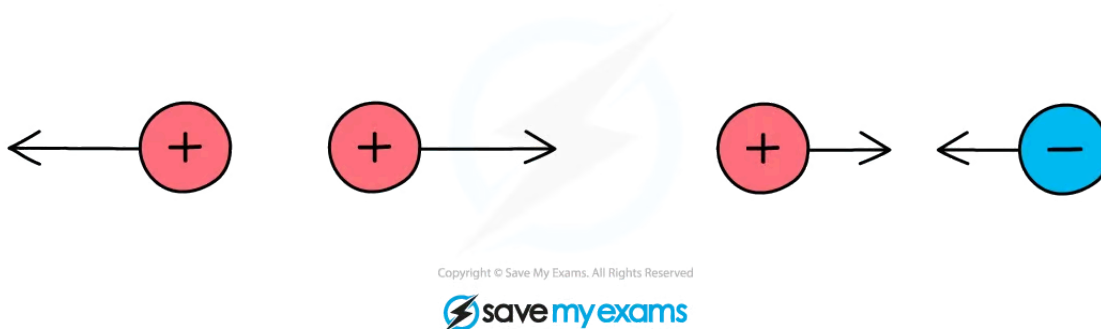


The water pushes up on the boat, this is the force of upthrust

▪ Electrostatic force

- There is an **electrostatic** force between two objects with **charge**
- **Like charges repel** one another, and **opposite charges attract** one another
- When an electron gets close to a positively charged ion, the ion exerts a pull force on the electron (**attraction**)
- When an electron gets close to another electron, the electrons experience a push force from one another (**repulsion**)

Electrostatic forces acting on charged particles



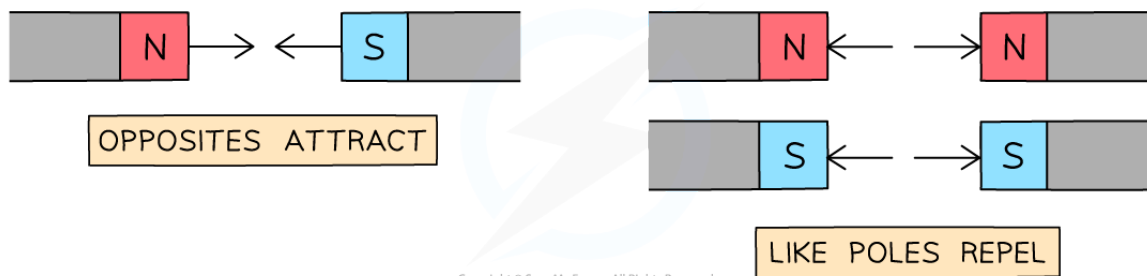
Like charges experience a repulsive push force and opposite charges experience an attractive pull force

▪ Magnetic force

- There is a **magnetic** force between objects with **magnetic poles**
- **Like poles repel** one another, **opposite poles attract** one another

- When a north pole gets close to a south pole, they experience a pull force from one another (**attraction**)
- When a north pole gets close to a north pole, they experience a push force from one another (**repulsion**)

Magnetic forces acting on magnets

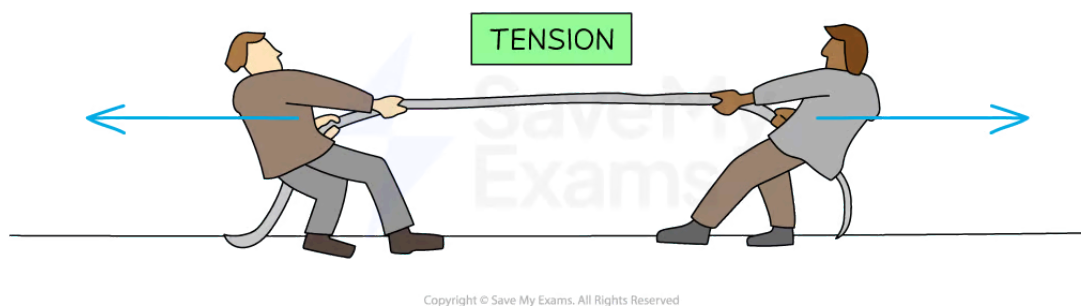


Opposite poles experience an attractive pull force, and like poles experience a repulsive push force

Tension

- Tension occurs in an object (like a rope or spring) that is **stretched**
- When a **pull** force is exerted on **each end** of an object, **tension** acts across the length of the object
- When two people pull a rope in opposite directions, tension acts along the rope and pulls back on each person

Tension force in a rope



Two people exert a pull force on the rope from each end, creating tension in the rope



Your notes

Examiner Tip

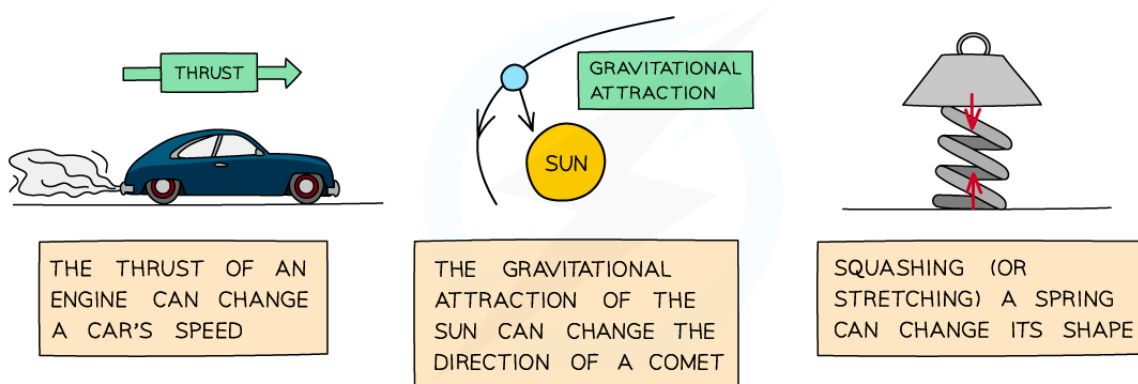
The force of gravitational attraction on an object is called its **weight**. Remember not to refer to this force as simply 'gravity', as this term can mean several different things, and examiners will probably mark it as wrong.

Similarly, when referring to **air resistance**, avoid using terms like 'wind resistance' (there is no such thing!) or 'air pressure', which is a different concept. **Drag** is an acceptable alternative to the force of air resistance because air resistance is a special type of drag.

Effects of forces

- When a force acts on an object, the force can affect the object in a variety of ways
- The object could:
 - change **speed**
 - change **direction**
 - change **shape**
- The effects of forces on an object often depend on the type of force acting
 - The push force (**thrust**) of an engine can cause a car to speed up, whilst the force exerted by the brakes (**friction**) can cause it to slow down
 - The gravitational pull of the Sun on a comet causes the comet to change direction
 - When two opposing forces push on each end of a spring, the spring changes shape (it **compresses**)

The effects of different forces on objects



Copyright © Save My Exams. All Rights Reserved



Forces can change the speed or direction of motion of an object, or even change its shape



Your notes

Scalar & Vectors

Scalar & vector quantities

- All quantities can be one of two types:
 - a **scalar**
 - a **vector**

Scalars

- Scalars are quantities that have **magnitude** but not direction
 - For example, **mass** is a **scalar** quantity because it has **magnitude** but no direction

Vectors

- Vectors are quantities that have both **magnitude** and **direction**
 - For example, **weight** is a **vector** quantity because it is a force and has **both** magnitude and direction

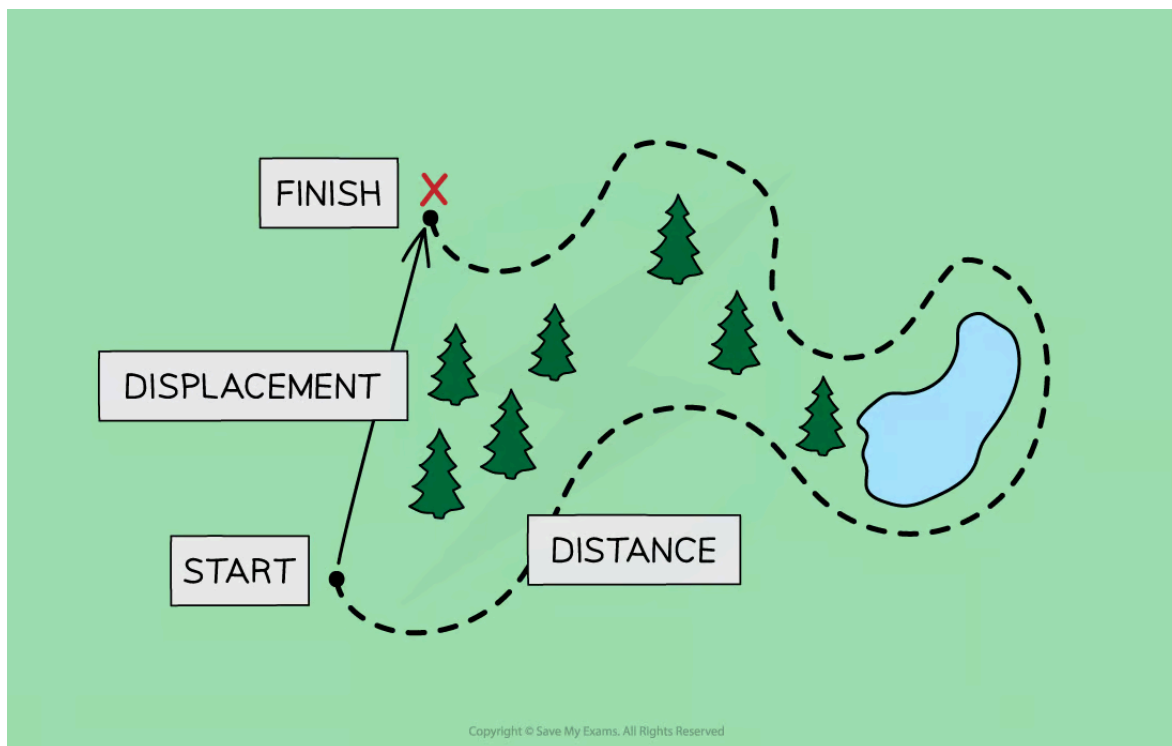
Distance and displacement

- Distance** is a measure of how far an object has travelled, **regardless of direction**
 - Distance is the total length of the path taken
 - Distance, therefore, has a **magnitude** but **no direction**
 - So, distance is a **scalar** quantity
- Displacement** is a measure of how far it is between two points in space, **including the direction**
 - Displacement is the **length** and **direction** of a **straight line** drawn from the starting point to the finishing point
 - Displacement, therefore, has a **magnitude** and a **direction**
 - So, displacement is a **vector** quantity

What is the difference between distance and displacement?



Your notes



Displacement is a vector quantity while distance is a scalar quantity

- When a student travels to school, there will probably be a **difference** in the distance they travel and their displacement
 - The **overall distance** they travel includes the total lengths of all the roads, including any twists and turns
 - The **overall displacement** of the student would be a straight line between their home and school, regardless of any obstacles, such as buildings, lakes or motorways, along the way

Speed and velocity

- Speed** is a measure of the **distance** travelled by an object per unit time, **regardless of the direction**
 - The speed of an object describes how fast it is moving, but **not** the direction it is travelling in
 - Speed, therefore, has **magnitude** but **no direction**
 - So, speed is a **scalar** quantity
- Velocity** is a measure of the **displacement** of an object per unit time, **including the direction**
 - The velocity of an object describes how fast it is moving **and** which direction it is travelling in
 - An object can have a **constant speed** but a **changing velocity** if the object is **changing direction**
 - Velocity, therefore, has **magnitude and direction**
 - So, velocity is a **vector** quantity

Examples of scalars & vectors

- The table below lists some common examples of scalar and vector quantities
- Corresponding vectors and their scalar counterparts are aligned in the table where applicable

Table of scalars and vectors



Your notes

Scalar	Vector
distance	displacement
speed	velocity
mass	weight
	force
	acceleration
	momentum
energy	
volume	
density	
temperature	
power	



Your notes

 Worked example

Astronaut A is in charge of training junior astronauts. For one of their sessions, they would like to explain the difference between mass and weight.

Suggest how Astronaut A should explain the difference between mass and weight, using definitions of scalars and vectors in your answer.

Answer:

Step 1: Recall the definitions of a scalar and vector quantity

- Scalar quantities have only a **magnitude**
- Vector quantities have both **magnitude and direction**

Step 2: Identify which quantity has magnitude only

- **Mass** is a quantity with **magnitude** only
- So mass is a **scalar** quantity
 - Astronaut A might explain to the junior astronauts that their mass will not change if they travel to outer space

Step 3: Identify which quantity has magnitude and direction

- **Weight** is a quantity with **magnitude and direction** (it is a force)
- So weight is a **vector** quantity
 - Astronaut A might explain to the junior astronauts that their weight will vary depending on their location in space

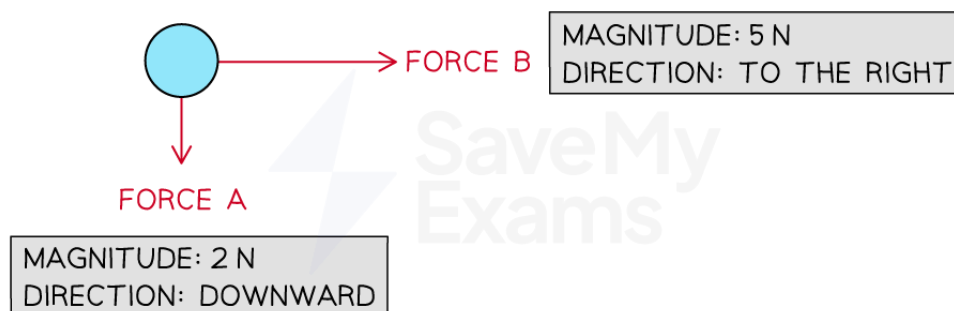


Your notes

Forces as Vectors

- **Vector** quantities can be represented by arrows
 - The **length** of the arrow represents the **magnitude**
 - The **direction** of the arrow indicates the **direction**
- **Force** is a **vector** quantity because force has **magnitude and direction**
- When using arrows to represent forces:
 - the **length** of the arrow represents the **magnitude** of the force
 - the **direction** of the arrow indicates the **direction** of the force
 - the **scale** of the arrows should be **proportional** to the relative magnitudes of the forces
 - an arrow for a 4 N force should be twice as long as an arrow for a 2 N force
 - the arrows should be labelled with the names of the forces, or a description of the forces
 - for example, weight, W, or the gravitational pull of the Earth on the object

Two forces acting on an object



Copyright © Save My Exams. All Rights Reserved

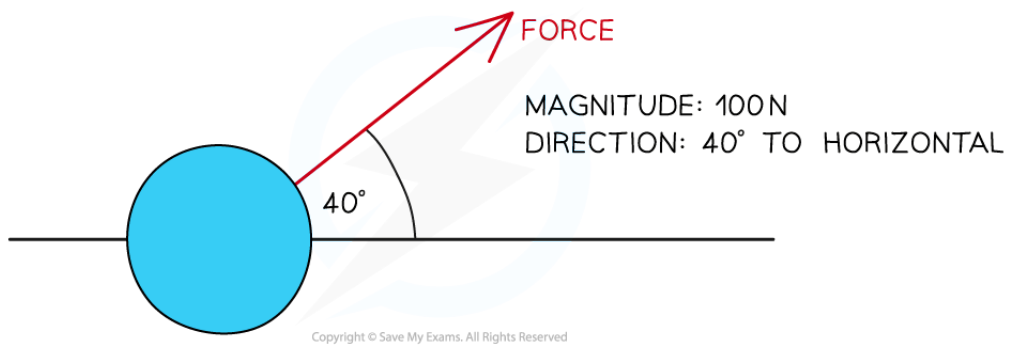
The length of the arrows are proportional to the magnitude of the forces, and show the direction that forces act in

- Not all forces are directed perfectly horizontally or vertically and so need to have an angle described for the direction
 - It is useful to describe an angle with respect to the vertical or the horizontal

A force acting at an angle



Your notes



A force of magnitude 100 N directed 40° from the horizontal



Your notes

Resultant Forces

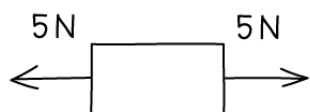
Calculating resultant force

What is a resultant force?

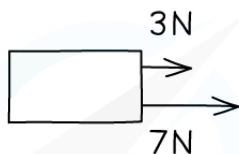
- A **resultant force** is a single force that describes all of the forces operating on a body
- When multiple forces act on one object, the forces can be combined to produce one net force that describes the **combined action** of all of the forces
- This single resultant force determines:
 - The **direction** in which the object will move as a result of all of the forces
 - The **magnitude** of the net force experienced by the object

Adding forces to find the resultant force

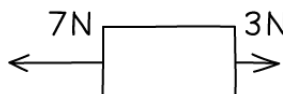
- Resultant forces can be calculated by adding all of the forces acting on the object
 - Forces working in opposite directions are **subtracted** from each other
 - Forces working in the same direction are **added** together
- If the forces acting in opposite directions are equal in size, then there will be no resultant force – the forces are said to be **balanced**



RESULTANT FORCE = 0N
(THE FORCES ARE
BALANCED)



RESULTANT FORCE =
 $3 + 7 = 10\text{ N}$
(TO THE RIGHT)



RESULTANT FORCE =
 $7 - 3 = 4\text{ N}$
(TO THE LEFT)

Copyright © Save My Exams. All Rights Reserved

Diagram showing the resultant forces on three different objects

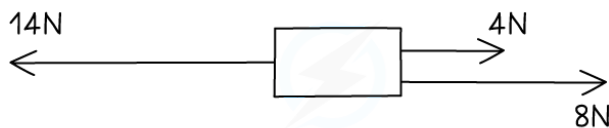
- Imagine the forces on the boxes as two people pushing on either side
 - In the first scenario, the two people are evenly matched – the box **doesn't move**
 - In the second scenario, the two people are pushing on the same side of the box, it moves to the right with their combined strength
 - In the third scenario, the two people are pushing against each other and are not evenly matched, so there is a **resultant force to the left**



Your notes

Worked example

Calculate the magnitude and direction of the resultant force in the diagram below.



Copyright © Save My Exams. All Rights Reserved

Answer:

Step 1: Add up all of the forces directed to the right

$$4\text{ N} + 8\text{ N} = 12\text{ N}$$

Step 2: Subtract the forces on the right from the forces on the left

$$14\text{ N} - 12\text{ N} = 2\text{ N}$$

Step 3: Evaluate the direction of the resultant force

- The force to the left is **greater** than the force to the right therefore the resultant force is directed to the left

Step 4: State the magnitude and direction of the resultant force

- The resultant force is 2 N to the left

Examiner Tip

When calculating resultant forces, always remember to provide **units** for your answer and to state whether the force is to the left, to the right, or maybe up or down. Always provide your final answer as a description of the **magnitude** and the **direction**, for example:

- Resultant Force = **4 N to the right**



Your notes

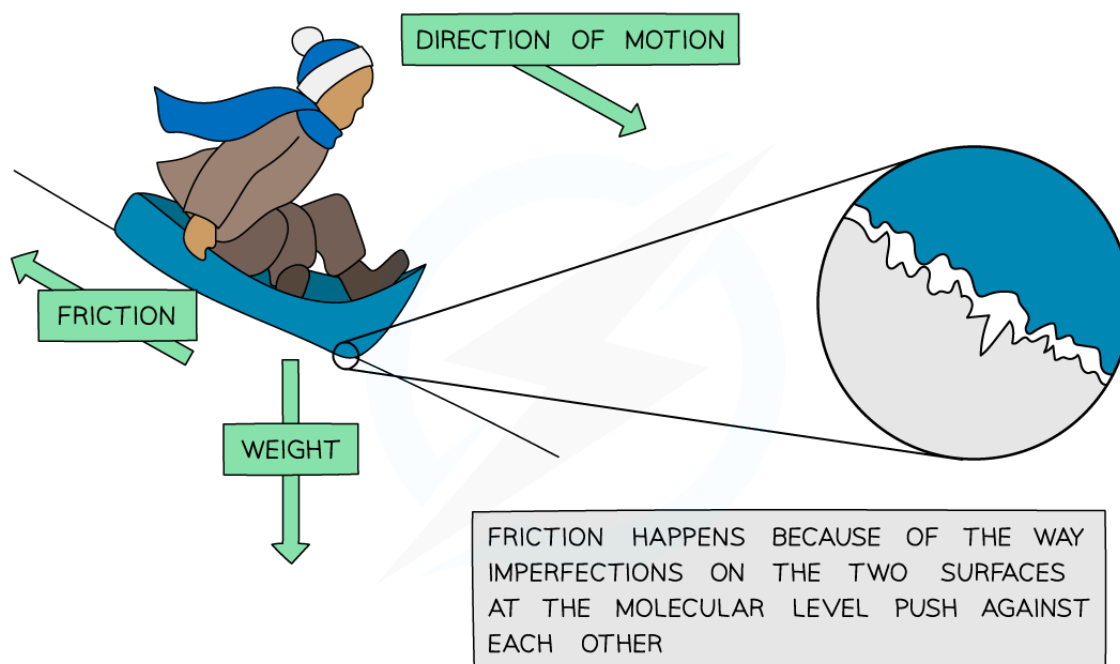
Friction

- Friction is defined as:

A force which opposes the motion of an object

- Frictional forces always act in the opposite direction to the object's motion
- Friction occurs when two (or more) surfaces rub against each other
 - At a molecular level, both surfaces contain **imperfections** - i.e. they are not perfectly smooth
 - These imperfections tend to push against each other

Friction acting between the surface of a sledge and the surface of the snow



Copyright © Save My Exams. All Rights Reserved

Friction is a force which opposes an objects motion, acting in the opposite direction to the motion of the object

 Examiner Tip

Remember from the revision note Types of Forces that drag force is a type of friction that acts to oppose the motion of objects moving through a fluid (a liquid or a gas).

Air resistance is a type of drag force, and therefore a type of friction, that acts to oppose the motion of objects moving through air.



Your notes



Your notes

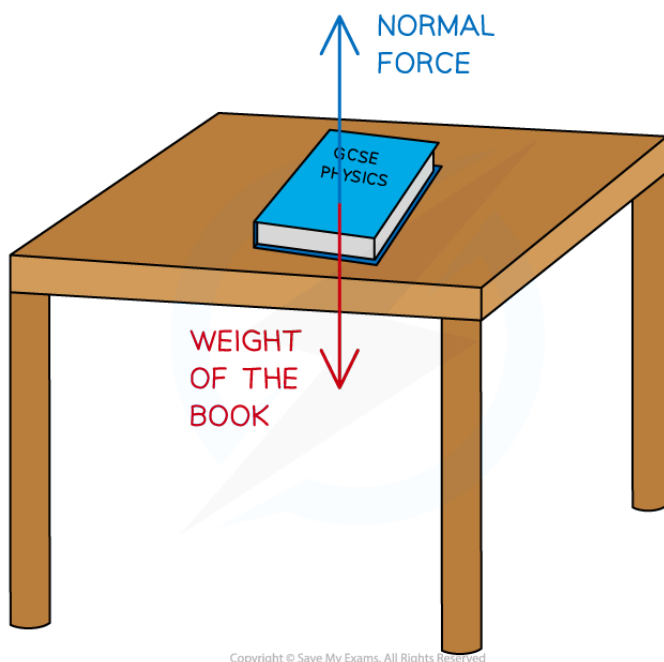
Unbalanced Forces

Unbalanced forces

- When **multiple forces** act on a **single object**, the forces can be added together to produce a **resultant force**
- When the forces acting on an object completely cancel out
 - the forces are **balanced**
 - the **resultant** force is **zero**
- When the forces acting on an object do not completely cancel out
 - the forces are **unbalanced**
 - there is a **resultant** force

Balanced forces

- **Balanced** forces mean that the forces have combined in such a way that they cancel each other out and no **resultant force** acts on the body
 - For example, the **weight** of a book on a desk is balanced by the **normal** force of the desk
 - As a result, **no resultant force** is experienced by the book, the book and the table are **equal** and **balanced**



Copyright © Save My Exams. All Rights Reserved

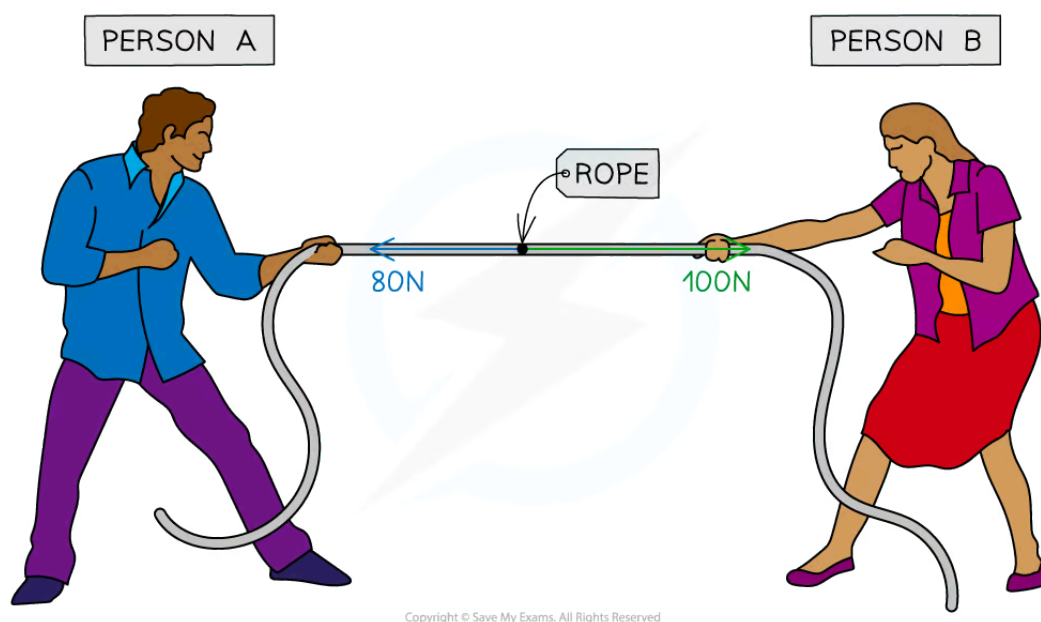
A book resting on a table is an example of balanced forces



Your notes

Unbalanced forces

- **Unbalanced** forces mean that the forces have combined in such a way that they do not cancel out completely and there is a **resultant force** on the object
 - For example, imagine two people playing a game of tug-of-war, working against each other on opposite sides of the rope
 - If person **A** pulls with 80 N to the left and person **B** pulls with 100 N to the right, these forces do not cancel each other out completely
 - Since person **B** pulled with more force than person **A** the forces will be unbalanced and the rope will experience a resultant force of 20 N to the right



Copyright © Save My Exams. All Rights Reserved

A tug-of-war is an example of when forces can become unbalanced

Unbalanced Forces, Mass & Acceleration

- When forces combine on an object in such a way that they do not cancel out, there is a **resultant force** on the object
- This resultant force causes the object to **accelerate** (i.e. change its velocity)
 - The object might **speed up**
 - The object might **slow down**
 - The object might **change direction**
- The relationship between resultant force, mass and acceleration is given by the equation:

$$F = m \times a$$

- Where:
 - F = resultant force, measured in newtons (N)
 - m = mass, measured in kilograms (kg)
 - a = acceleration, measured in metres per second squared (m/s^2)
- This equation is also known as Newton's second law of motion



Your notes



Your notes

 Worked example

A car salesperson claims that their best car has a mass of 900 kg and can accelerate from 0 to 27 m/s in 3 seconds.

Calculate:

- a) The acceleration of the car in the first 3 seconds.
- b) The force required to produce this acceleration.

Answer:

Part (a)**Step 1: List the known quantities**

- Initial velocity = 0 m/s
- Final velocity = 27 m/s
- Time, $t = 3$ s

Step 2: State the equation for acceleration

$$a = \frac{(v - u)}{t}$$

Step 3: Calculate the acceleration

$$a = \frac{(27 - 0)}{3}$$
$$a = 9 \text{ m/s}^2$$

Part (b)**Step 1: List the known quantities**

- Mass of the car, $m = 900$ kg
- Acceleration, $a = 9 \text{ m/s}^2$

Step 2: Identify which law of motion to apply

- The question involves quantities of **force**, **mass** and **acceleration**, so Newton's second law is required:

$$F = ma$$

Step 3: Calculate the force required to accelerate the car

$$F = 900 \times 9$$

$$F = 8100 \text{ N}$$



Your notes



Your notes

 Worked example

A passenger of mass 70 kg travels in a car at a speed of 20 m/s. The vehicle is involved in a collision, which brings the car (and the passenger) to a halt in 0.1 seconds.

Calculate:

- a) The deceleration of the car (and the passenger).
- b) The decelerating force on the passenger.

Answer:

Part (a)

Step 1: List the known quantities

- Initial velocity, $u = 20 \text{ m/s}$
- Final velocity, $v = 0 \text{ m/s}$
- Time, $t = 0.1 \text{ s}$

Step 2: Write out the equation for acceleration

$$a = \frac{(v - u)}{t}$$

Step 3: Calculate the deceleration of the car (and the passenger):

$$a = \frac{(0 - 20)}{0.1}$$

$$a = \frac{-20}{0.1}$$

$$a = -200 \text{ m/s}^2$$

Part (b)

Step 1: List the known quantities

- Mass of the passenger, $m = 70 \text{ kg}$
- Acceleration (deceleration, in this case), $a = -200 \text{ m/s}^2$

Step 2: State the relationship between resultant force, mass and acceleration

- This question involves quantities of force, mass and acceleration, so the appropriate equation for this case is:

$$F = ma$$

Step 3: Calculate the decelerating force

$$F = 70 \times (-200)$$

$$F = -14\,000 \text{ N}$$



Your notes

Examiner Tip

Remember that the resultant **force** is a **vector** quantity. Examiners may ask you to comment on **why** its value is negative - this happens when the resultant force acts in the **opposite direction** to the object's motion. In the worked example above, the resultant force **opposes** the passenger's motion, slowing them down (**decelerating** them) to a halt, this is why it has a minus symbol.



Your notes

Weight

Weight

- Weight is defined as:

The force experienced by an object with mass when placed in a gravitational field

Weight and mass

- Weight** and **mass** are **different** in physics
- Mass** is a measure of how much **matter** there is in an object
 - Mass has **magnitude** but not direction
 - Therefore, mass is a **scalar** quantity
- Weight** is a force
 - Forces have **magnitude and direction**
 - Therefore, weight is a **vector** quantity

Gravitational field strength

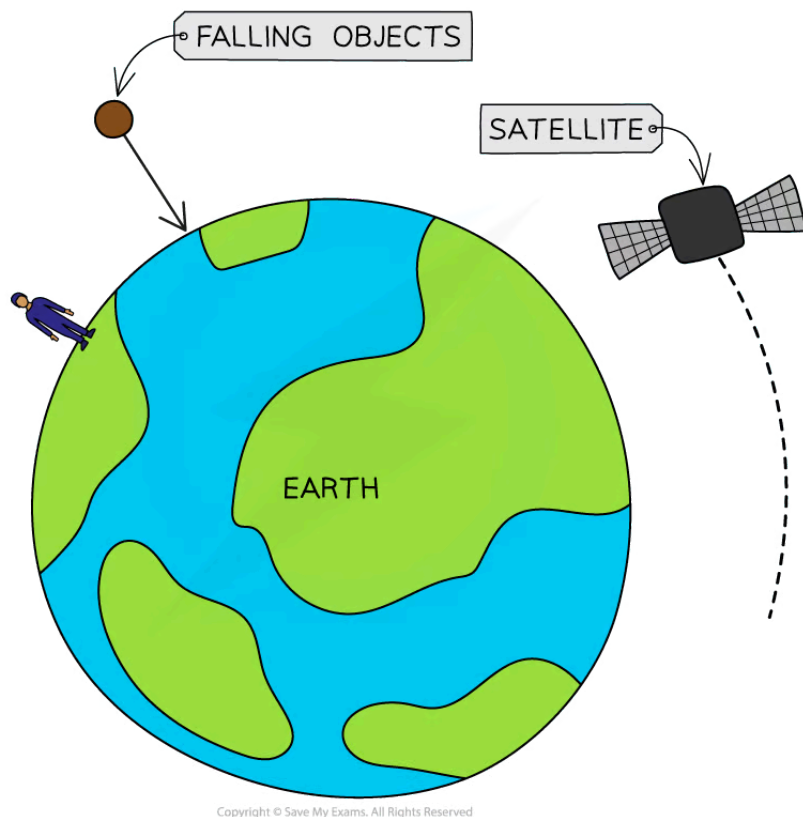
- Planets have strong gravitational fields
 - Hence, they attract nearby masses with a strong gravitational force
- Different planets have different gravitational field strengths
 - This depends on the mass of the planet
 - More massive planets have stronger gravitational field

Weight, mass and gravitational field strength

- Because of weight:
 - Objects stay firmly on the ground
 - Objects will always fall to the ground
 - Satellites are kept in orbit



Your notes



Some of the phenomena associated with gravitational attraction and the weight force

- Weight, mass and gravitational field strength are related using the equation:

$$W = mg$$

- Where:
 - W = weight, measured in newtons (N)
 - m = mass, measured in kilograms (kg)
 - g = gravitational field strength, measured in newtons per kilogram (N/kg)
- The **gravitational field strength** on Earth is **10 N/kg**
- g is also used to describe the acceleration of an object in freefall in a gravitational field
 - The **acceleration of freefall** on Earth is **10 m/s²**
 - These quantities are two ways of describing the thing
- The weight that an object experiences depends on:
 - The **object's** mass
 - The mass of the **planet** attracting it
- The mass of an object and the weight acting on it are **directly proportional**

- If one doubles, the other also doubles
- If one is halved, the other is also halved
- The **magnitude** of the weight force depends on the **gravitational field strength**



Your notes

Worked example

NASA's Artemis mission aims to send the first woman astronaut to the Moon. Isabelle hopes to one day become an astronaut. She has a mass of 40 kg.

Earth's gravitational field strength is 10 N/kg, and the Moon's gravitational field strength is 2 N/kg.

Discuss the difference between Isabelle's weight on Earth, and her weight on the Moon.

Answer:

Step 1: State the equation linking weight and mass

- The equation linking weight and mass is:

$$W = m \times g$$

Step 2: List the known values

- Earth's gravitational field strength, $g = 10 \text{ N/kg}$
- The Moon's gravitational field strength, $g = 2 \text{ N/kg}$
- Mass, $m = 40 \text{ kg}$

Step 3: Calculate Isabelle's weight on Earth

- Substituting the values of mass and Earth's gravitational field strength into the equation gives:

$$W = 40 \times 10 = 400 \text{ N}$$

Step 4: Calculate Isabelle's weight on the Moon

- Substituting the values of mass and the Moon's gravitational field strength into the equation gives:

$$W = 40 \times 2 = 80 \text{ N}$$

Step 5: Discuss the two values of weight

- Isabelle's weight is **greater** on Earth than on the Moon
- This is because the Earth has a **larger** gravitational field strength than the Moon, so Isabelle's weight force (the force of gravity pulling down on her) is larger on Earth than on the Moon

 Examiner Tip

It is a common misconception that mass and weight are the same, but they are in fact **very different**

- Since weight is a **force** - it is a **vector** quantity
- Since mass is an **amount** - it is a **scalar** quantity

You do not need to remember the value of g on Earth; it will be given to you in the exam.



Your notes



Your notes

Stopping Distance

Stopping distance

- The **stopping distance** of a car is defined as:
The stopping distance of a car is the total distance travelled during the time it takes to stop in an emergency
- The stopping distance is the sum of the distance travelled as the driver makes the decision to stop plus the distance travelled as the driver applies the brakes

Stopping distance formula

- The stopping distance is calculated using the following formula:
Stopping distance = Thinking distance + Braking distance

Thinking distance

- Thinking distance** is defined as
Thinking distance is the distance travelled in the time it takes the driver to react to an emergency and prepare to stop
- The **main factors** affecting thinking distance are:
 - The **speed** of the car
 - The **reaction time** of the driver
- The **reaction time** is defined as:
A measure of how much time passes between seeing something and reacting to it
- The average reaction time of a human is 0.25 s
- Reaction time is increased by:
 - Tiredness**
 - Distractions** (e.g. using a mobile phone)
 - Intoxication** (i.e. consumption of **alcohol** or **drugs**)

Braking distance

- Braking distance** is defined as
the distance travelled under the braking force in metres (m)
- For a given braking force, the **greater the speed** of the vehicle, the **greater the stopping distance**

Calculating stopping distance

- For a given braking force, the **speed** of a vehicle determines the size of the **stopping distance**
- The **greater the speed** of the vehicle, the **larger the stopping distance**

The Effect of speed on stopping distance

THINKING DISTANCE

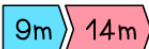
BRAKING DISTANCE

AVERAGE CAR LENGTH = 4 METRES (13 FEET)



Your notes

20 mph
(32 km/h)  = 12 METRES (40 FEET)
OR THREE CAR LENGTHS

30 mph
(48 km/h)  = 23 METRES (75 FEET)
OR SIX CAR LENGTHS

40 mph
(64 km/h)  = 36 METRES (118 FEET)
OR NINE CAR LENGTHS

50 mph
(80 km/h)  = 53 METRES (175 FEET)
OR THIRTEEN CAR LENGTHS

60 mph
(96 km/h)  = 73 METRES (240 FEET)
OR EIGHTEEN CAR LENGTHS

70 mph
(112 km/h)  = 96 METRES (315 FEET)
OR TWENTY-FOUR CAR LENGTHS

Copyright © Save My Exams. All Rights Reserved

A vehicle's stopping distance increases with speed. At a speed of 20 mph the stopping distance is 12 m, whereas at 60 mph the stopping distance is 73 m (reproduced from the UK Highway Code)

A Table Showing Speed and Stopping Distance

Speed (mph)	Speed (m/s)	Stopping Distances (m)
20	9	12
30	14	23
40	18	36
50	22	53
60	27	73



Your notes

 Worked example

At a speed of 20 m/s, a particular vehicle had a stopping distance of 40 metres. The car travelled 14 metres whilst the driver was reacting to the incident in front of him.

What was the braking distance?

- A 54 m
- B 34 m
- C 26 m
- D 6 m

ANSWER: C

Step 1: Identify the different variables

- Stopping distance = 40 m
- Thinking distance = 14 m

Step 2: Rearrange the formula for stopping distance

$$\text{Stopping distance} = \text{Thinking distance} + \text{Braking distance}$$

$$\text{Braking distance} = \text{Stopping distance} - \text{Thinking distance}$$

Step 3: Calculate and identify the correct braking distance

$$\text{Braking distance} = 40 - 14$$

$$\text{Braking distance} = 26 \text{ metres}$$

- Therefore, the answer is **C**



Your notes

Factors affecting Stopping Distance

- There are various factors which can affect a vehicle's stopping distance
 - **Vehicle speed**
 - The greater the speed, the greater the vehicle's braking distance will be
 - This is because the brakes will need to do more work to bring the vehicle to a stop
 - **Vehicle mass**
 - The more massive the vehicle, the more distance it will travel as it comes to a stop
 - **Road conditions**
 - Wet or icy roads make the brakes less effective and the vehicle travels further as it comes to a stop
 - **Driver reaction time**
 - Thinking distance is increased if the driver is distracted, for example by a phone, satnav, radio or a person
 - Thinking distance is increased if the driver is tired, on certain types of medication, or is under the influence of alcohol or drugs



Your notes

Terminal Velocity

Terminal velocity

- **Terminal velocity** is the fastest speed that an object can reach when falling
- Terminal velocity is reached when the upward and downward acting **forces** are **balanced**
 - The **resultant force** on the object reaches **zero**
 - The object no longer accelerates and a **constant** terminal **velocity** is reached

Falling objects

- Falling objects experience **two forces**:
 - **Weight**
 - **Air resistance**
- The force of air resistance **increases** as the object's **speed** increases
- This is because the object collides with air particles as it moves through the air
 - The faster the object is travelling, the more collisions it has with the air particles
- The weight of the object does not change
- This is because $W = mg$
 - The mass, m , of the object does not change
 - The acceleration of freefall, g , does not change

Reaching terminal velocity

Skydiver in freefall reaching terminal velocity



The skydiver initially accelerates downward due to the force of weight. The upward force of air resistance increases as they fall until eventually it equals the weight force and terminal velocity is reached

- At the **instant** the skydiver steps out of the plane, the support force of the plane is no longer acting on the skydiver, but they are not yet falling, so the only force exerted on them is the **weight** force
 - There is a downward acting **resultant force** on the skydiver
 - The resultant force is equal to the weight force
 - The skydiver accelerates downward at **maximum acceleration**
- As the skydiver begins to **fall**, the force of air resistance is very **small** because the skydiver's speed is small
 - There is a downward acting **resultant force** on the skydiver
 - The resultant force is equal to the weight force **minus** the force of air resistance
 - The skydiver accelerates downward but the **acceleration decreases**
- As the skydiver **accelerates**, their speed increases, so the force of **air resistance increases**
 - There is a downward acting **resultant force** on the skydiver
 - The resultant force is equal to the weight force minus the force of air resistance
 - The skydiver accelerates downward but the **acceleration continues to decrease**

- As the skydiver's **acceleration decreases**, their speed increases at a slower and slower rate
 - Eventually, the skydiver reaches a speed at which the force of air resistance is **equal** to the force of weight
 - The forces are **balanced**, so the **resultant force is zero**
 - The skydiver no longer accelerates and a **constant velocity** is reached
 - This is **terminal velocity**



Your notes



Your notes

 Worked example

A small object falls out of an aircraft. Choose words from the list to complete the sentences below:

Friction Gravity Air pressure

Accelerates Falls at a steady speed Slows down

- (a) The weight of an object is the force of _____ which acts on it.
- (b) When something falls, initially it _____.
- (c) The faster it falls, the larger the force of _____ which acts on it.
- (d) Eventually it _____ when the force of friction equals the force of gravity acting on it.

Answer:

Part (a)

- The weight of an object is the force of **gravity** which acts on it.
 - The weight force is due to the Earth's gravitational pull on the object, so weight is due to gravity

Part (b)

- When something falls, initially it **accelerates**.
 - The **resultant force** on the object is **very large** initially, so it accelerates
 - This is because there is a **large unbalanced force** downwards (its weight) – the upward force of air resistance is very small to begin with

Part (c)

- The faster it falls, the larger the force of **friction** which acts on it.
 - The force of **air resistance** is due to **friction** between the object's motion and **collisions with air particles**
 - Air particles try to slow the object down, so air itself produces a **frictional force**, called air resistance (sometimes called **drag**)

Part (d)

- Eventually it **falls at a steady speed** when the force of friction equals the force of gravity acting on it.
 - When the upwards **air resistance** grows enough to **balance** the downwards **weight** force, the resultant force on the object is **zero**
 - This means the object isn't accelerating – rather, it is moving at a **steady** (terminal) **speed**



Your notes

 Examiner Tip

The force of gravity on an object is called its **weight**. If you are asked to name this force, use this word: don't call it 'gravity', as this term could also mean gravitational field strength, and so might be marked wrong. Additionally, remember to identify **air resistance** as the upward force on a falling object. This force **gets larger** as the object speeds up, but the weight of the object stays constant. Don't confuse 'air resistance' with 'air pressure' as these are two different concepts!



Your notes

Core Practical: Investigating Force & Extension

Core practical 2: investigating force & extension

Experiment 1: investigating force and extension for springs and rubber bands

- The aim of this experiment is to investigate the relationship between **force** and **extension** for a spring and a rubber band:

Variables

- Independent variable = Force, F
- Dependent variable = Extension, e

Equipment

Equipment list

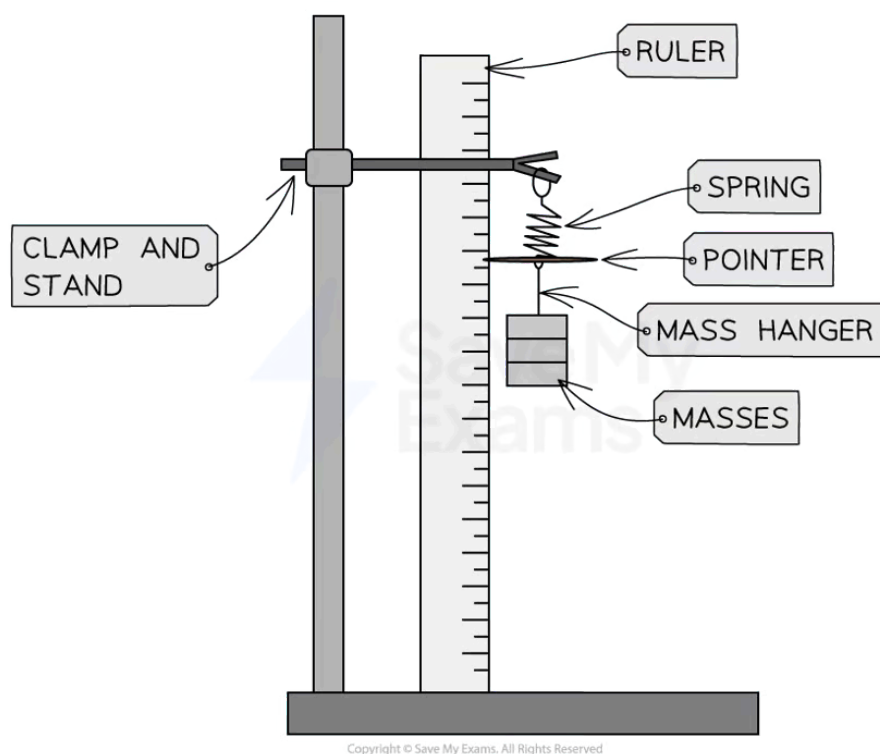
Equipment	Purpose
Clamp and stand	To apply an upward force to the spring and rubber band
Ruler	To measure original length and extension
Spring and rubber band	To measure the extension of
5 × 100 g masses	To apply a downward force to the spring and rubber band
100 g mass hanger	To hold additional masses
Pointer (also called a fiducial marker)	To accurately read the extension from the ruler
G-clamp	To secure the clamp stand to the bench so that the equipment does not fall over

- Resolution of measuring equipment:
 - Ruler = 1 mm

Method



Your notes



Example set-up of the equipment used to investigate force and extension for a spring

1. Align the marker to a value on the ruler with no mass added, and record this initial length of the spring / rubber band
2. Add the 100 g mass hanger onto the spring / rubber band
3. Record the mass (in kg) and position (in cm) from the ruler now that the spring / rubber band has extended
4. Add another 100 g to the mass hanger
5. Record the new mass and position from the ruler now that the spring / rubber band has extended further
6. Repeat this process until all masses have been added
7. Remove the masses and repeat the entire process again, until it has been carried out a total of three times, and an average length (for each mass attached) is calculated

Example results table



Your notes

MASS/kg	FORCE/N	LENGTH 1/m	LENGTH 2/m	LENGTH 3/m	AVERAGE LENGTH/m	EXTENSION/m
0						
0.1						
0.2						
0.3						
0.4						
0.5						
0.6						

Copyright © Save My Exams. All Rights Reserved

Experiment 2: investigating force and extension for metal wires

- The aim of this experiment is to investigate the relationship between **force** and **extension** for a metal wire

Variables

- Independent variable = Force, F
- Dependent variable = Extension, e

Equipment

Equipment list

Equipment	Purpose
Ruler	To measure original length and extension
Spring, rubber band and metal wire	To measure the extension of
5 × 100 g masses	To apply a downward force to the wire
100 g mass hanger	To hold additional masses
Tape (to use as a marker on the wire)	To accurately read the extension from the ruler

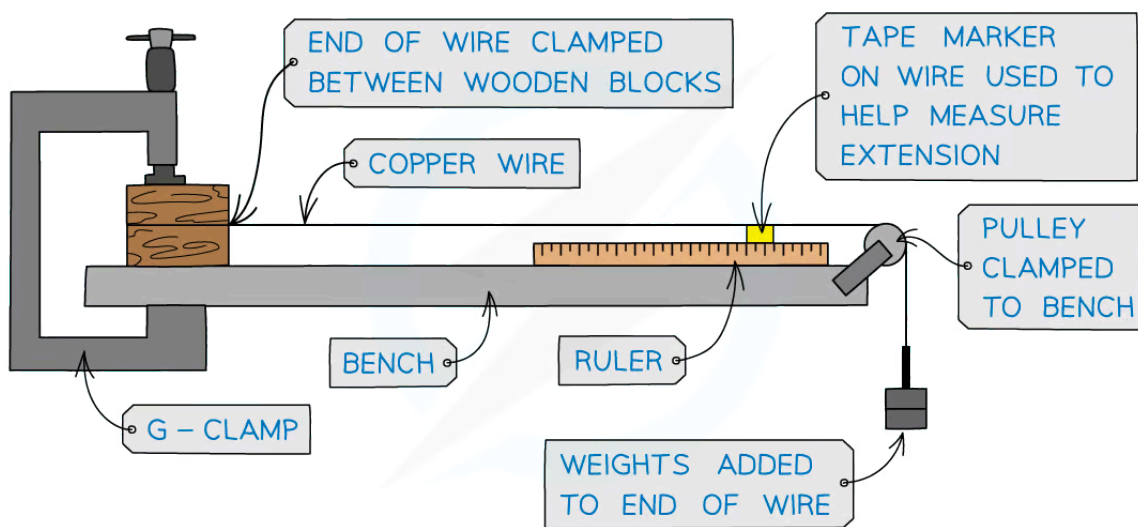


Your notes

Bench pulley	To allow the masses to hang vertically from the metal wire
G-clamp & wooden blocks	To apply an opposing force on the end of the wire

- Resolution of measuring equipment:
 - Ruler = 1 mm

Method

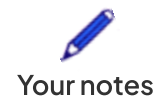


Copyright © Save My Exams. All Rights Reserved

Example set-up of equipment to investigate the force and extension for a metal wire

- Set up the apparatus so the wire is taut with no masses added
- Measure the original length of the wire using a metre ruler and mark a reference point with tape preferably near the beginning of the scale eg. at 1 cm
- Record the initial length of the wire to the marker
- Add a 100 g mass onto the mass hanger
- Read and record the new reading of the tape marker from the meter ruler now that the metal wire has extended
- Repeat this process until all masses have been added
- Remove the masses and repeat the entire process again, until it has been carried out a total of three times, and an average length (for each mass attached) is calculated

Example results table



$F = mg$

$e = \text{NEW MARKER READING} - \text{REFERENCE POINT READING}$

MASS / kg	FORCE / N	NEW MARKER READING / m	EXTENSION / m
0.1			
0.2			
0.3			
0.4			
0.5			

Copyright © Save My Exams. All Rights Reserved

Analysis of results

- The force, F added to the spring / rubber band / metal wire is the **weight** of the mass
- The weight is calculated using the equation:

$$W = m \times g$$

- Where:
 - W = weight in newtons (N)
 - m = mass in kilograms (kg)
 - g = gravitational field strength on Earth in newtons per kg (N/kg)
- Therefore, multiply each mass by gravitational field strength, g , to calculate the force, F
 - The force can be calculated by multiplying the mass (in kg) by 10 N/kg
- The **extension e** of the spring / rubber band is calculated using the equation:

$$e = \text{average length} - \text{original length}$$

- The final length is the length of the spring / rubber band recorded from the ruler after the masses were added
- The **extension e** of the metal wire is calculated using the equation:

$$e = \text{new marker reading} - \text{reference point reading}$$

- The original length is the length of the spring / rubber band / metal wire when there were **no** masses attached

1. Plot a graph of the force against extension for the spring / rubber band / metal wire

2. Draw a line or curve of best fit
3. If the graph has a **linear** region (is a straight line), then the force is **proportional** to the extension



Your notes

Evaluating the experiments

Systematic errors

- Make sure the measurements on the ruler are taken at eye level to avoid **parallax error**

Random errors

- The accuracy of such an experiment is improved with the use of a pointer (a fiducial marker)
- Wait a few seconds for the spring / rubber band / metal wire to fully extend when a mass is added, before taking the reading for its new length
- Make sure to check whether the spring has not gone past its limit of proportionality otherwise, it has been stretched too far

Safety considerations

- Wear goggles during this experiment in case the spring, rubber band or wire snaps
- Stand up while carrying out the experiment, making sure no feet are directly under the masses
- Place a mat or a soft material below the masses to prevent any damage in case they fall
- Use a G clamp to secure the clamp stand to the desk so that the clamp and masses do not fall over
 - As well as this, place each mass carefully on the hanger and do not pull the spring so hard that it breaks or pulls the apparatus over



Examiner Tip

Remember - for the spring and rubber band, the extension measures how much the object has stretched by and can be found by **subtracting the original length from each of the subsequent lengths**.

For the metal wire, each extension is measured by finding the difference between the **new marker point** and the **original reference point**.

A common mistake is to calculate the increase in length by each time instead of the total extension. If each of your extensions is roughly the same, then you might have made this mistake!



Your notes

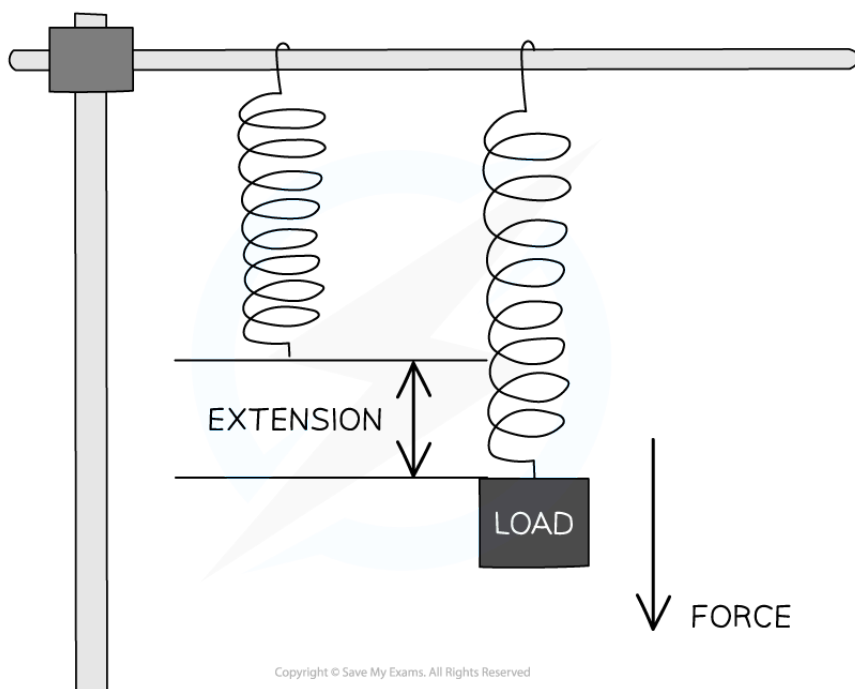
Hooke's Law

Hooke's law

- The relationship between the extension of an elastic object and the applied force is defined by **Hooke's Law**
- Hooke's Law states that:

The extension of an elastic object is directly proportional to the force applied, up to the limit of proportionality

- Directly proportional means that as the force is increased, the extension increases
 - If the force is doubled, then the extension will double
 - If the force is halved, then the extension will also halve
- The **limit of proportionality** is the point beyond which the relationship between force and extension is no longer directly proportional
 - This limit varies according to the material



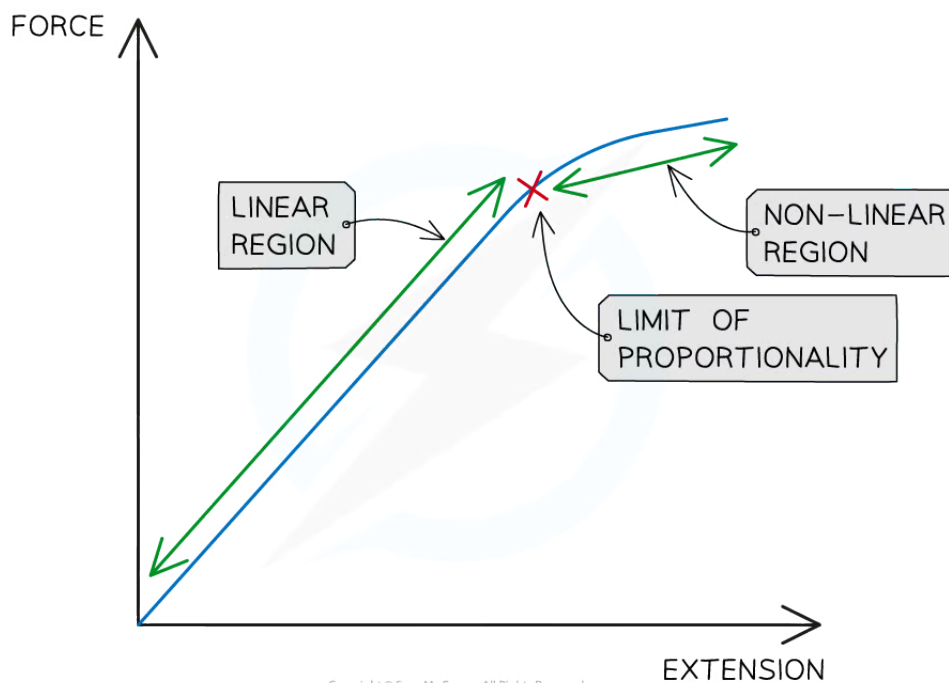
Hooke's Law states that a force applied to a spring will cause it to extend by an amount proportional to the force



Your notes

The force-extension graph

- Hooke's law is the **linear** relationship between force and extension
 - This is represented by a **straight line** on a force-extension graph
- Any material beyond its limit of proportionality will have a non-linear relationship between force and extension



Copyright © Save My Exams. All Rights Reserved

Hooke's Law is associated with the linear region of a force-extension graph. Beyond the limit of proportionality, Hooke's law no longer applies



Your notes

Elastic Behaviour

- **Elastic behaviour** is the ability of a material to **recover its original shape** after the forces causing the deformation have been removed
- **Deformation** is a change in the original shape of an object
- Deformation can be either:
 - elastic
 - inelastic

Elastic Deformation

- **Elastic deformation** is when the object **does** return to its original shape after the deforming forces are removed
- Elastic deformation results in a change in the object's shape that is **not permanent**
- Examples of materials that undergo elastic deformation are:
 - Rubber bands
 - Fabrics
 - Steel springs

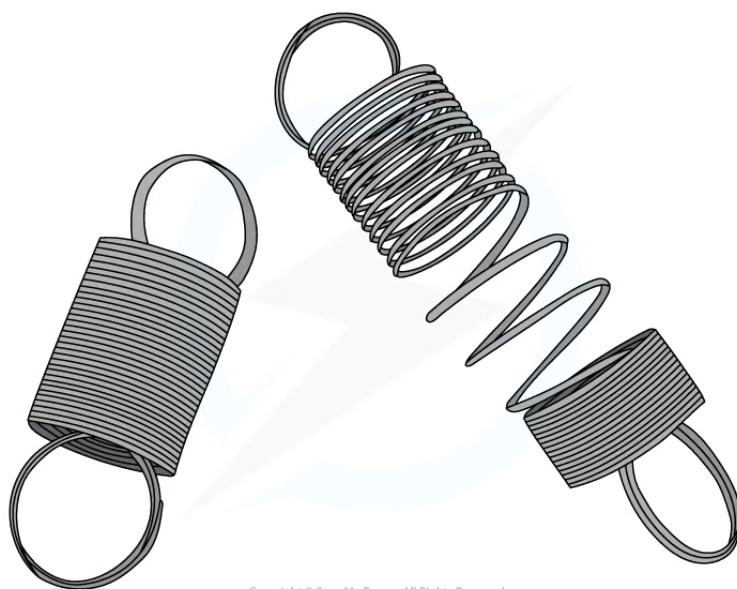
Inelastic Deformation

- **Inelastic deformation** is when the object **does not** return to its original shape after the deforming forces are removed
- Inelastic deformation results in a change in the object's shape that is **permanent**
- Examples of materials that undergo inelastic deformation are:
 - Plastic
 - Clay
 - Glass

Elastic behaviour of a spring



Your notes



Copyright © Save My Exams. All Rights Reserved

The spring on the right has undergone inelastic deformation, it's shape has been permanently deformed